Anupam Bisht, PhD

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EDUCATION

2021-2024	**Doctor of Philosophy** Department of Biomedical Engineering, University of Calgary, Calgary, Canada Thesis: Devices and computational methods for preclinical mechanistic deep brain stimulation studies in freely moving rodents. Supervisor: Dr. Kartikeya Murari Doctoral Committee: Dr. Zelma Kiss, Dr. Patrick Whelan
2019-2021	**Master of Science in Biomedical Engineering (Transferred to PhD)** Department of Biomedical Engineering, University of Calgary, Calgary, Canada
2015-2019	**Bachelor of Technology in Electrical and Electronics Engineering** Vellore institute of Technology (VIT), Vellore, India.

POSITIONS AND SCIENTIFIC APPOINTMENT

Feb 2025-Present	Postdoctoral research associate, Dr. Elizabeth Hillman's Lab, Department of Imaging Sciences**,** St. Jude Children's Research Hospital, Memphis, Tennessee, United States
Jan – April, 2024	Sessional Instructor, Department of Electrical and Software Engineering, University of Calgary, Calgary, Alberta, Canada
Sept 2019-Dec 2025	Graduate student, Dr. Kartikeya Murari's Lab, Department of Biomedical Engineering, University of Calgary, Calgary, Alberta, Canada
Dec-April,2019	Undergraduate final year research project student, Dr. Ramasubba Reddy, Indian Institute of Technology, Madras, Chennai, India
May-June, 2018	Summer research student, Dr. Aditya Murthy, Indian institute of Science (IISc), Bangalore, India
Feb 2017-April 2019	Undergraduate research student, Dr. Geethanjali P, Vellore Institute of Technology, Vellore, India.

JOURNAL PUBLICATIONS

1. **A Bisht**, C Badenhorst, Z Kiss, P Whelan, K Murari., "Deep brain stimulation of A13 region evokes robust locomotory response in rats", Journal of Neurophysiology 2025 133:5, 1594-1606

2. **A Bisht**, G Peringod, L Yu, N Cheng, G Gordon, K Murari., "Tetherless miniaturized point detector device for monitoring widefield cortical surface hemodynamics in mice", J Biomed Opt. 2025 Feb;30(Suppl 2):S23904. doi: 10.1117/1.JBO.30.S2.S23904. Epub 2025 Mar 19. PMID: 40110227; PMCID: PMC11922257.

3. R L’Orsa, **A Bisht**, L Yu, K Murari, D T. Westwick, G R. Sutherland, and K J. Kuchenbecker “Reflectance from in-bore optical fibers enhances needle puncture detection” Front Robot AI. 2025 May 6;12:1429327. doi: 10.3389/frobt.2025.1429327. PMID: 40395363; PMCID: PMC12090360.

4. **A Bisht**, K Simone, J S. Bains, K Murari, Distinguishing motion artifacts during optical fiber-based in-vivo hemodynamics recordings from brain regions of freely moving rodents, Neurophotonics, 11(S1), S11511 (2024), doi: 10.1117/1.NPh.11.S1.S11511

5. **A Bisht,** S Srivastava, G Purushothaman, et al. “A New 360° Rotating Type Stimuli for Improved SSVEP Based Brain Computer Interface.” Biomedical Signal Processing and Control, vol. 57, Mar. 2020, p. 101778. ScienceDirect, https://doi.org/10.1016/j.bspc.2019.101778.

CONFERENCE PROCEEDINGS

1. **A Bisht,** M Omidi, A Hosseini, and K Murari. "A Data-Driven Approach for Extracting Oxygen Saturation From in-Vivo Fiber Photometry Data", OPTICA Biophotonics congress, April 2025

2. R L’Orsa, **A Bisht**, L Yu, K Murari, D T. Westwick, G R. Sutherland, and K J. Kuchenbecker “Reflectance outperforms force and position in model-free needle puncture detection.” 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Orlando, FL, USA, 2024, pp. 1-7, doi: 10.1109/EMBC53108.2024.10782536.*

3. **A Bisht**, G Peringod, G R. Gordon, J S. Bains, K Murari, "System for simultaneous measurement of fluorescence and hemodynamics (SSMFH) from deep brain regions of freely moving rodents," Proc. SPIE 12828, Neural Imaging and Sensing 2024, 128280A (12 March 2024); https://doi.org/10.1117/12.3002776

4. **A Bisht**, L Yu, N Cheng, K Murari, "TinyIOMS: A Wireless Miniaturised System for Monitoring Hemodynamics from Surface Brain Regions," *2023 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, Toronto, ON, Canada, 2023, pp. 1-5, doi: 10.1109/BioCAS58349.2023.10388820.

5. M. Namazi, G. Peringod, **A. Bisht**, J. Bains, G. R. Gordon and K. Murari, "Silicon Photomultiplier-based Low-light in vivo Fiber Photometry," *2023 IEEE Biomedical Circuits and Systems Conference (BioCAS)*, Toronto, ON, Canada, 2023, pp. 1-5, doi: 10.1109/BioCAS58349.2023.10389054.

6. **A Bisht,** K Simone, G Peringod, J Bains, G Gordon, K Murari, “Investigating Presence of Motion Artifacts in the Oxygen Saturation Signal during In-Vivo Fiber Photometry.” Neural Imaging and Sensing 2022, edited by Qingming Luo et al., SPIE, 2022, p. 8. DOI.org (Crossref), https://doi.org/10.1117/12.2609877.

CONFERENCE ABSTRACTS

1. **A Bisht**, C Badenhorst, I Acosta, Z Kiss, P Whelan, K Murari, "Deep Brain Stimulation (DBS) of A13 region in rats using a wireless DBS device", Neurohike Conference**,** 16th November 2023, Kananaskis, Alberta, Canada.

2. T Fuzesi, N Rasiah, W Nicola, J Bains, M R Carvajal, T Chomiak, D Rosenegger, **A Bisht**, K Simone, L Molina, G Petrie, M Hill, K Murari, "Hypothalamic persistent activity states evoked by stress", CNS*2023, Leipzig, Germany

3. T Fuzesi, N Rasiah, W Nicola, J Bains, M R Carvajal, T Chomiak, D Rosenegger, **A Bisht**, K Simone, L Molina, G Petrie, M Hill, K Murari , "Hypothalamic persistent activity states evoked by stress", CAN 2023, Montreal, Canada

PRESENTATIONS/INVITED TALKS

July 2024	Hillman lab, Zuckerman Mind Brain Behavior Institute, Columbia University, NY, USA
June 2024	Hotchkiss Brain Institute Research Day, University of Calgary, Calgary. (Selected trainee talk)
May 2024	Howard Hughes Medical Institute (HHMI), Janelia farms, Ashburn, Virginia, USA (Invited Talk)
May 2024	W.M Keck Center for Cellular Imaging, University of Virginia, Charlottesville, USA (Invited Seminar)

TEACHING (at University of Calgary, Canada)

1. 2020-2023 **Electronic Devices and Materials (ENEL 361, Integrated Learning Stream)**
   - Marking exams**,** debugging hardware circuits (online during the COVID-19 pandemic) in lab activities
2. 2022 **Professional Development (ENEL 300, Integrated Learning Stream)**
   - Involving students in group discussions during class activities.
   - Providing periodic project feedback and evaluation.
3. 2020-2022 **Circuits II (ENEL 343, Integrated Learning Stream)**
   - Marking exams**,** debugging hardware circuits (online during the COVID-19 pandemic) in lab activities.
4. 2020 **Digital Circuits (ENEL 353)**
   - Marking exams, evaluating labs.
5. 2021 **Biomedical Signals, Systems and Instrumentation I (BMEN 388)**
   - Helped prepare a lab for the new course.

FELLOWSHIPS, AWARDS AND RECOGNITIONS

2024	Richard Motyka Graduate Scholarship in Engineering, University of Calgary
2024	Hotchkiss Brain Institute Conference Travel Award, University of Calgary
2024	Graduate student association professional development grant, University of Calgary
2023	SPIE Photonics West 2024 Student Conference Support Grant

2023	2023 IEEE Circuits and Systems Student Travel Grant
2020-2022	2022 BME Research Award, University of Calgary
2022	ESE Research Week graduate student presentation award, University of Calgary
2022	Robert B. Paugh Memorial Scholarship in Engineering, University of Calgary
2020-2021	Alberta Graduate Excellence Scholarship, University of Calgary
2021	Biomedical Engineering Travel Award, University of Calgary
2019	Rank 2 out of 250 students in the Class of 2019 of the Electrical and Electronics Engineering program at Vellore Institute of Technology, Vellore, India
2018	Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR) summer fellowship
2015-2018	Merit Scholarship for excellent academic performance, Vellore Institute of Technology, Vellore, India

PROFESSIONAL SERVICE

- Reviewer for Journal of Biomedical Signal Processing and Control

PROFESSIONAL MEMBERSHIPS

- SPIE
- IEEE
- OPTICA

VOLUNTEER AND OUTREACH ACTIVITIES

In Canada

- Volunteered at a workshop at Discovery Day organized by the Canadian Medical Hall of Fame "Biomedical Engineers: What do they do and how do you get into this field?" on 18th October 2023. We reached out to about 20 students.
- Volunteered as a Judge for the Calgary Youth Science Fair, held at the University of Calgary, Canada, 21st April 2023. Here I reviewed 5 projects from students from grades 6-10, along with 2 other judges.
- Mentor for an incoming international graduate student at the International Student Mentorship Program, University of Calgary, Canada (August-December 2020).
- Presenter for 2 sessions at East Brook Elementary School, Brooks, Alberta, Canada as part of an outreach program of Calgary Optics and Photonics Society. We reached out to about 150 students in total during both the sessions. (October 2019). Here I presented demonstrations to explain optics concepts.

In India:

- Visted Kulethi primary school Champawat, Uttarakhand, India and interacted with 10 students (Grade 7) working at Atal Tinkering Lab, Champawat, Uttarakhand. Provided suggestions to improve outreach activities and usage of equipment (such as 3-D printers, Arduino Uno, etc) to the current Trainer at the lab. (Nov 20-22, 2023).
- A volunteer at Kulethi primary school Champawat, Uttarakhand, India. I helped to promote tablet assisted teaching in the primary school. I also helped in video making of basic science experiments in Hindi, which are uploaded in YouTube. (Link, June 2016).

REFERENCES

Reference -1 (PhD Advisor)
Dr. Kartik Murari, University of Calgary
Email: kmurari@ucalgary.ca

Reference -2 (Doctoral Committee)
Dr. Patrick Whelan, University of Calgary
Email: whelan@ucalgary.ca

Reference -3 (Doctoral Committee)
Dr. Zelma Kiss, University of Calgary
Email: zkiss@ucalgary.ca

Reference-4 (Postdoctoral Advisor)
Dr. Elizabeth Hillman, St. Jude Children's Research Hospital
Email: elizabeth.hillman@stjude.org